

A
Mini Project Report
on
**ML Based Transaction Pattern Analysis for Anti-Money
Laundering Compliance.**

Submitted in partial fulfillment of the requirements for the degree
Third Year Engineering – Computer Science Engineering (Data Science)

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ABSTRACT

This project focuses on developing an ML-based transaction pattern analysis system for Anti-Money Laundering (AML) compliance, enhancing financial security and fraud detection. By leveraging advanced machine learning algorithms, the system identifies suspicious transaction patterns, assigns risk scores, and generates detailed compliance reports. Techniques such as anomaly detection, clustering, and supervised learning models improve the accuracy of fraud detection, minimizing false positives.

The platform will feature a user-friendly interface for seamless integration with financial systems, allowing real-time monitoring and reporting. By automating compliance processes and improving risk assessment, this project aims to strengthen AML efforts, ensuring regulatory adherence while reducing financial crime risks efficiently.

Chapter 1

Introduction

Money laundering remains a major challenge for financial institutions, enabling illicit activities such as fraud, terrorism financing, and organized crime. Traditional rule-based AML systems often fall short in detecting sophisticated fraudulent patterns, leading to inefficiencies and high false positive rates. As financial transactions grow in volume and complexity, there is a pressing need for intelligent solutions that can analyze vast amounts of data, identify anomalies, and enhance compliance with regulatory standards.

This project introduces an AI-driven AML website that leverages machine learning to monitor financial transactions, detect suspicious activities, and ensure regulatory compliance. By utilizing advanced techniques such as anomaly detection, clustering, and predictive modeling, the system improves detection accuracy while minimizing false alarms. Additionally, the platform automates reporting and risk assessment, providing financial institutions with a proactive approach to combating money laundering and safeguarding financial integrity.

The system can also learn from new data to stay up-to-date with changing fraud tactics. It can be easily scaled and adapted as needed to handle new types of financial crimes. By combining machine learning and automation, this platform will make it easier for financial institutions to fight against money laundering and protect their systems.

1.1. Purpose:

The purpose of this project is to develop an AI-driven AML system to enhance fraud detection and regulatory compliance. Using machine learning, it improves accuracy, automates risk assessment, and strengthens transaction analysis.

- **Fraud Detection:** Detecting fraudulent transactions is essential to preventing financial crimes such as money laundering. By analyzing transaction patterns, machine learning models can identify suspicious activities, such as unusually large transfers, rapid successive transactions, or dealings with high-risk entities. Flagging these transactions in real-time allows financial institutions to take immediate action, reducing the risk of financial fraud.

- **Risk Assessment:** Assigning risk scores to transactions helps prioritize investigations and prevent high-risk activities. By evaluating transaction history, spending behavior, and anomalies, machine learning models can classify transactions based on their likelihood of fraud. This risk-based approach enables financial institutions to make informed decisions on whether to approve, review, or block transactions.
- **Anomaly Detection:** Machine learning algorithms can effectively identify deviations from normal transaction behaviors, allowing for proactive fraud prevention. Techniques like Isolation Forest and Random Forest help detect unusual activities, such as sudden spikes in transaction amounts or patterns inconsistent with a user's typical behavior. Early detection of these anomalies enhances security measures and reduces financial risks.
- **User-Friendly Interface:** A well-designed dashboard is crucial for efficient fraud monitoring and reporting. An intuitive interface provides real-time insights, visualizations of transaction patterns, and alerts for suspicious activities. By simplifying complex data analysis, financial analysts and investigators can quickly assess risks, generate reports, and take appropriate actions to mitigate fraud.

1.2. Problem Statement

Money laundering poses a significant challenge to financial institutions, facilitating illegal activities such as fraud, corruption, and terrorism financing. Traditional rule-based AML systems are often ineffective in detecting evolving money laundering techniques, leading to high false positives, missed threats, and increased compliance costs. As financial transactions grow in volume and complexity, manual monitoring becomes impractical, making fraud detection inefficient and reactive rather than proactive.

- **Money laundering enables illegal activities and challenges financial institutions.** Money laundering facilitates criminal activities like fraud, corruption, organized crime, and terrorism financing by concealing the origins of illicit funds. Financial institutions face serious challenges in detecting these activities, as criminals exploit loopholes in financial systems to move and legitimize illegal money.
- **Traditional rule-based AML systems are ineffective and generate high false positives.**

Conventional AML systems rely on static rules and thresholds that cannot adapt to sophisticated laundering methods. These systems often trigger large numbers of false positives, flagging legitimate transactions unnecessarily while still missing complex fraud patterns. This results in wasted resources, delayed responses, and increased compliance costs.

- **The increasing volume and complexity of transactions make manual monitoring impractical.**

With the rise of digital payments, cross-border transactions, and cryptocurrencies, the financial ecosystem has grown complex and fast-moving. Manual monitoring is no longer scalable, leading to reactive fraud detection and delayed action, which makes financial institutions vulnerable to money laundering activities.

- **A machine learning approach strengthens financial security and operational efficiency.**

By integrating machine learning into AML systems, financial institutions can shift from reactive to proactive fraud detection strategies. This approach improves security, ensures regulatory compliance, reduces operational costs, and minimizes manual intervention, thereby safeguarding the financial ecosystem against money laundering risks.

1.3. Objectives:

The primary objective of this project is to develop a machine learning-driven Anti-Money Laundering (AML) system that enhances fraud detection. Traditional AML methods often struggle with identifying sophisticated money laundering patterns, leading to inefficiencies and high false positives. By leveraging risk score analysis, transaction monitoring, and automated report generation, this system aims to provide a more effective and adaptive solution for financial institutions.

- **Analyze financial transactions to detect unusual patterns using machine learning using isolation forest algorithm:** Financial fraud detection relies on advanced machine learning techniques to analyze transactions, identify unusual patterns, and assess risk levels. One of the key approaches is using the Isolation Forest algorithm to detect anomalies in financial transactions. This method works by isolating outliers more efficiently than traditional clustering techniques. By randomly partitioning data and building decision trees, the algorithm identifies fraudulent transactions based on how

quickly they are isolated. Transactions that significantly deviate from normal behavior, such as sudden large withdrawals, frequent small transactions, or purchases from high-risk locations, can be flagged for further investigation. This real-time detection capability helps financial institutions mitigate fraud risks before significant losses occur.

- **Assign risk scores based on transaction history, demographics, and high-risk connections using random forest algorithm:** Another crucial aspect of fraud detection is assigning risk scores to transactions based on historical patterns, user demographics, and associations with high-risk entities. The Random Forest algorithm is well-suited for this task as it evaluates multiple factors to generate a reliable risk assessment. By analyzing transaction history, the model identifies deviations in spending behavior, frequency, and amounts. Demographic factors such as age, occupation, and geographical location are also considered to determine the likelihood of fraudulent activity. Additionally, the algorithm examines connections between accounts, flagging users associated with previously identified fraudulent entities. The ensemble learning nature of Random Forest ensures high accuracy and reduces overfitting, making risk scores more precise and actionable. Financial institutions can use these scores to prioritize investigations, approve or reject transactions, and enhance overall fraud prevention strategies.
- **Create Suspicious Activity Reports for flagged transactions:** Once suspicious transactions are identified, they need to be documented in Suspicious Activity Reports (SARs) for regulatory compliance and further examination by financial authorities. SARs provide a detailed breakdown of flagged transactions, including key details such as transaction amount, date, and involved parties. They also outline the reasons for flagging, such as detected anomalies, high-risk connections, or unusual behavioral patterns. Advanced machine learning models can automate the generation of SARs by summarizing key insights and risk factors, allowing fraud investigators to focus on high-priority cases. These reports play a critical role in anti-money laundering (AML) efforts, ensuring that financial institutions remain compliant with legal requirements while actively combating fraudulent activities.

1.4. Scope:

The scope of this project focuses on developing a machine learning-driven Anti-Money Laundering (AML) system that enhances fraud detection, risk assessment, and compliance monitoring. By utilizing advanced ML algorithms, the system efficiently analyzes financial transactions, detects suspicious activities, and automates report generation. Designed for banks, financial institutions, and regulatory bodies, this system ensures improved accuracy, reduced false positives, and streamlined compliance processes.

- **Transaction Monitoring:** By continuously analyzing financial activities, the system detects anomalies and suspicious patterns in real time, enabling proactive fraud prevention.
- **Risk Score Analysis:** The system assigns risk scores to transactions based on historical patterns, user demographics, and connections to high-risk entities, helping financial institutions prioritize investigations.
- **Report Generation:** The system creates detailed compliance reports, summarizing flagged transactions and risk assessments, assisting financial institutions in regulatory reporting.
- **Adaptive Learning:** The system continuously refines its fraud detection models by learning from new transaction data, evolving fraud tactics, and user interactions over time.

Chapter 2

Literature Review

A literature review provides an analysis of existing research, offering insights into methodologies, outcomes, and gaps in the current knowledge. A literature review serves as a foundational component of academic research, providing a comprehensive analysis of previous studies related to a specific topic. This chapter presents an extensive review of scholarly research and industry studies focusing on AI-driven AML systems. It critically examines the methodologies, data sources, and analytical models used in recent literature. The reviewed literature not only sheds light on the technical advancements in the field but also explores the broader implications for regulatory compliance, data privacy, and cross-border financial monitoring.

In the context of AI-driven Anti-Money Laundering (AML) systems, multiple studies have explored how AI and machine learning techniques can enhance fraud detection and financial security by identifying suspicious transactions and improving risk assessment. The focus is on understanding how artificial intelligence (AI), including machine learning (ML) and deep learning (DL), contributes to enhancing fraud detection, identifying suspicious activities, and improving risk scoring in financial systems. As financial crimes continue to grow in complexity and scale, traditional rule-based AML systems are struggling to keep pace with the dynamic and evolving patterns of money laundering. Moreover, this review identifies the current limitations, challenges, and future research directions in the domain.

- 1. Anti-Money Laundering Systems: A Systematic Literature Review by A. Alsuwailem and A. Saudagar (2020):** This study reviewed 27 research papers published between 2015 and 2020, categorizing AML techniques based on AI models, data sources, and evaluation methods. Alsuwailem and Saudagar highlighted how AI can enhance fraud detection but also pointed out limitations in real-world implementation, such as data quality issues, lack of standard benchmarks, and challenges in regulatory compliance. The review emphasized the need for adaptive, explainable models to keep up with evolving money laundering tactics and improve practical effectiveness.

2. **Network Analytics for Anti-Money Laundering by B. Deprez, T. Vanderschueren, B. Baesen's, T. Verdonck, and W. Verbeke (2024):** Deprez and colleagues explored the use of network analytics and graph neural networks (GNNs) in AML, showcasing their ability to uncover hidden connections in transactional data. By using the Elliptic dataset for evaluation, they demonstrated that GNNs significantly outperform traditional AML methods in detecting suspicious activity. The study emphasized modeling relational data and transaction networks to improve prediction accuracy and suggested further research into temporal and dynamic graph models.
3. **Machine Learning and Deep Learning Integration in AML by N. Husnaningtyas, G. Hanin, T. Dewayanto, and M. Malik (2023):** This paper reviewed 26 studies from 2018 to 2023, focusing on the effectiveness of integrating machine learning and deep learning techniques in AML. The authors found that hybrid AI models consistently delivered better detection accuracy. Decision trees were noted as commonly used models, and the paper recommended combining multiple AI methods, including ensemble learning, to build more robust AML systems capable of adapting to new fraud patterns.
4. **AML and Anti-Terrorism Financing: A Survey by N. Al-Suwaidi and H. Nobanee (2020):** Al-Suwaidi and Nobanee focused on the regulatory and policy aspects of AML and anti-terrorism financing. They identified legal gaps that limit the implementation of AI in financial monitoring and stressed the need for international cooperation and standardized laws. Their study called for stronger regulations and the integration of advanced technologies like AI and blockchain to enhance the effectiveness of AML frameworks globally.

The literature indicates that AI techniques, particularly machine learning, deep learning, and graph-based analytics, have shown promising results in enhancing AML systems. While hybrid and network-driven approaches offer high accuracy, challenges related to data quality, interpretability, and regulation persist. Future research must focus on improving explainability, leveraging federated learning for secure data sharing, and developing globally adaptable AML solutions.

Chapter 3

Proposed System

The proposed ML-driven Anti-Money Laundering (AML) system aims to strengthen financial security by detecting suspicious transactions and reducing false positives. Traditional rule-based AML systems often generate excessive false alerts, leading to inefficiencies in fraud detection. To address this, the system will leverage machine learning algorithms such as Isolation Forest for anomaly detection and Random Forest for risk score generation. By analyzing transaction patterns, historical trends, and predefined risk factors, the system will effectively flag unusual activities while minimizing false alarms. This ensures that high-risk transactions receive immediate attention, allowing financial institutions to prioritize investigations and improve fraud detection accuracy.

In addition to fraud detection, the system will provide transaction monitoring and automated reporting to enhance compliance with regulatory frameworks. Continuous transaction surveillance will help identify suspicious patterns as they occur, enabling proactive fraud prevention. When a high-risk transaction is detected, the system will automatically generate detailed reports, including risk assessments and transaction histories, to assist compliance officers in decision-making. By integrating intelligent monitoring, risk-based scoring, and automated reporting, the proposed AML system will streamline compliance processes, enhance fraud prevention, and create a more secure financial ecosystem.

3.1. Features and Functionality:

The proposed ML-driven anti-money laundering (AML) system is designed to enhance financial security by leveraging machine learning techniques for detecting suspicious transactions. By analyzing transaction patterns, assigning risk scores, and providing real-time monitoring, the system aims to improve fraud detection while minimizing false positives. Additionally, it will incorporate adaptive learning to refine detection accuracy over time. The following features outline the key functionalities of the system:

1. **Suspicious Transaction Detection:** The AI-driven AML system will utilize machine learning models to analyze vast amounts of financial transaction data, identifying patterns and anomalies that indicate potential money laundering activities. By examining transaction history, frequency, and unusual fund movements, the system can

differentiate between normal financial behavior and suspicious activities. Techniques such as Isolation Forest will be employed to detect outliers, ensuring that fraudulent transactions are flagged accurately. This proactive approach reduces manual oversight and enhances the efficiency of fraud detection.

2. **Risk-Based Scoring:** Each financial transaction will be assessed based on historical data, predefined risk factors, and learned patterns to generate a risk score. Using Random Forest algorithms, the system will classify transactions into different risk levels—low, medium, or high—based on indicators such as transaction amount, frequency, location, and counterparties involved. High-risk transactions will be prioritized for further investigation, ensuring that compliance teams focus their efforts on the most critical threats. This scoring mechanism minimizes false positives while improving the accuracy of financial crime detection.
3. **Transaction Monitoring:** The system will operate in real-time, continuously tracking and analyzing transaction behaviors across financial networks. By leveraging machine learning models, it will detect deviations from normal transaction patterns, such as sudden large fund transfers, rapid successive transactions, or unusual international fund flows. Automated alerts will be generated when transactions exceed certain risk thresholds, enabling financial institutions to act swiftly and prevent fraudulent activities before they escalate. This real-time monitoring approach enhances security, regulatory compliance, and fraud prevention.
4. **Automated Reporting:** To streamline compliance and regulatory oversight, the system will generate automated reports whenever a suspicious transaction is detected. These reports will include detailed transaction histories, associated risk scores, and key risk indicators, providing compliance officers with actionable insights. By reducing manual effort in report generation, financial institutions can respond quickly and efficiently to potential threats. Additionally, these reports can be integrated into regulatory frameworks, ensuring that financial institutions meet AML compliance requirements while maintaining an audit trail for future reference.

Chapter 4

Requirement Analysis

Requirement analysis is a systematic process that involves identifying, documenting, and evaluating the needs and expectations of stakeholders. This phase defines the system's scope, functionalities, and constraints, ensuring that the final product aligns with user requirements. By thoroughly understanding these needs, the development team can create a system that effectively detects suspicious transactions, minimizes false positives, and ensures regulatory compliance.

- **Dataset:** The system requires a structured dataset containing financial transactions, customer profiles, and historical fraud cases. This dataset will be used to train and test machine learning models for accurate fraud detection.
- **Risk Scoring Mechanism:** The system should assign risk scores to transactions based on factors such as transaction amount, frequency, and involved entities, enabling financial institutions to focus on high-risk activities.
- **Anomaly Detection Engine:** A machine learning-powered module should analyze transaction patterns, identify outliers, and detect deviations from normal financial behavior to minimize fraudulent activities.
- **Transaction Monitoring Module:** A real monitoring system is necessary to track financial transactions, detect unusual activities, and flag suspicious transactions for further review.
- **Reporting System:** The system should generate suspicious activity reports automatically, providing compliance officers with detailed insights into flagged transactions.
- **Data Preprocessing and Feature Engineering:** The system should incorporate data cleaning, transformation, and feature extraction techniques to enhance model performance and improve fraud detection accuracy.

Chapter 5

Project Design

Project design refers to the structured approach of defining the architecture, components, and workflow of a system to ensure it meets its intended objectives. It involves outlining the technical and functional aspects of the project, including system architecture, data flow, and user interactions. A well-defined project design serves as a blueprint for implementation, ensuring efficiency, scalability, and alignment with stakeholder requirements. By focusing on both high-level conceptualization and detailed planning, project design minimizes development risks and enhances the overall effectiveness of the system.

5.1. Use Case Diagram:

A use case diagram is a graphical representation of the interactions between users (actors) and a system, depicting the various functionalities the system provides. It is a key component of the Unified Modeling Language (UML) and helps in understanding the system's functional requirements. By illustrating the relationships between different actors and use cases, it provides a clear overview of how the system is expected to operate, ensuring better design and development planning.

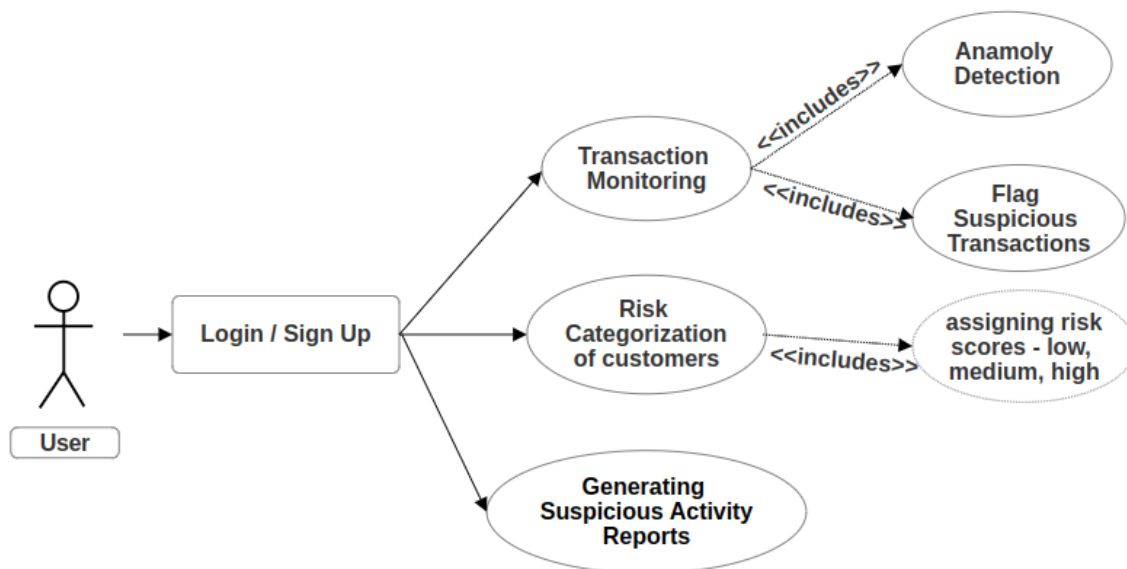


Fig. 5.1.1 Use Case Diagram.

Figure 5.1.1 represents the Use Case Diagram for an Anti-Money Laundering (AML) system, illustrating the interactions between a user and key system functionalities. The system enables Transaction Monitoring, which detects anomalies and flags suspicious transactions, and Risk Categorization, which assigns customers a risk score (low, medium, high) based on their transaction behavior. If suspicious activities are identified, the system generates Suspicious Activity Reports (SARs) for regulatory compliance and further investigation. Following are the key features:

- **User Login / Sign Up:** It Allows authorized users (bank officials, compliance officers) to access AML functionalities and Ensures security and role-based access to transaction monitoring and risk assessment features.
- **Transaction Monitoring:** It Continuously scans financial transactions to detect suspicious activities and Uses rule-based and AI-driven models to identify irregularities in transaction patterns. It includes Anomaly Detection – Detects unusual activities like rapid high-value transactions or structured deposits and Anomaly Detection – Detects unusual activities like rapid high-value transactions or structured deposits.
- **Risk Categorization of Customers:** It Assigns customers a low, medium, or high risk score based on their transaction behavior. Risk scoring helps prioritize investigations and prevent unnecessary manual reviews. It includes Assigning Risk Scores – Customers with frequent flagged transactions are categorized as high-risk and Behavioral Analysis – Monitors long-term patterns to assess if a customer's risk score should change.
- **Generating Suspicious Activity Reports:** It Compiles flagged transactions and risk categorizations into reports for regulatory compliance and Helps financial institutions and government bodies investigate and take action against financial crimes.

Relationship Analysis

- User Login/Sign Up is the entry point to access AML features securely.
- Transaction Monitoring is central to detecting financial crime and works with Risk Categorization to evaluate customers.

- Flagged transactions and risk scores influence SAR generation, which is essential for reporting to regulators.
- The system continuously learns from flagged transactions, refining its risk categorization models.

5.2. DFD (Data Flow Diagram):

A DFD is a graphical representation of the flow of data through a system. It shows the processes that transform data, the data stores that hold data, and the data flows that connect them.

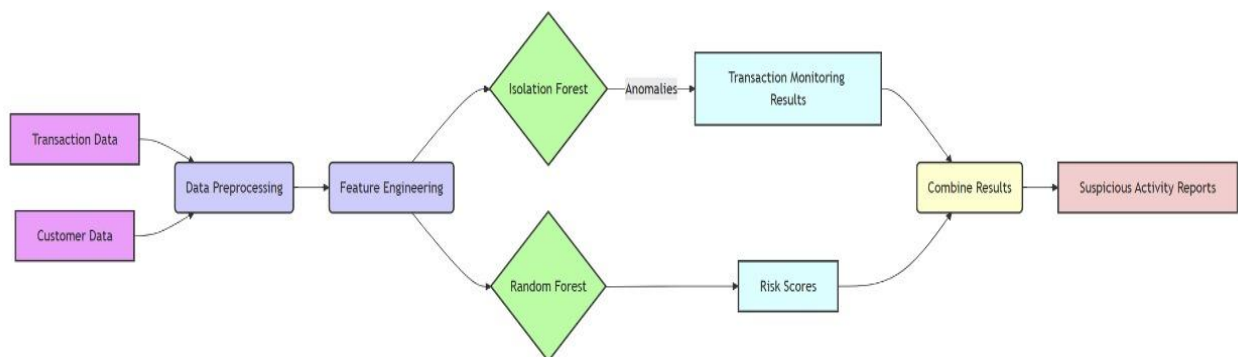


Fig. 5.2.1 Data Flow diagram

The above Fig. 5.2.1 Data Flow Diagram (DFD) represents an ML Based Transaction Pattern Analysis for Anti-Money Laundering Compliance that detects suspicious financial activities. The process begins with Transaction Data (financial transactions) and Customer Data (user profiles and history), which undergo Data Preprocessing to clean and prepare the information. Next, Feature Engineering extracts relevant attributes like transaction frequency and risk factors. The system then applies Isolation Forest to identify anomalies, which are sent to Transaction Monitoring Results, while Random Forest assigns Risk Scores based on past fraudulent behavior. These results are then combined to determine if a transaction is suspicious. Finally, flagged transactions are recorded in Suspicious Activity Reports (SARs) for further investigation by financial institutions and regulators. This system enhances fraud detection by leveraging anomaly detection and risk scoring techniques.

5.3. System Architecture:

System architecture is the fundamental structure of a system that defines its components, their interactions, and the overall workflow. It serves as a blueprint that guides the design and implementation of a system, ensuring that all elements work together efficiently to achieve a specific goal. In the case of ML Based Transaction Pattern Analysis for Anti-Money Laundering Compliance, the architecture outlines how transaction data is collected, processed, analyzed, and reported to detect and prevent financial crimes.

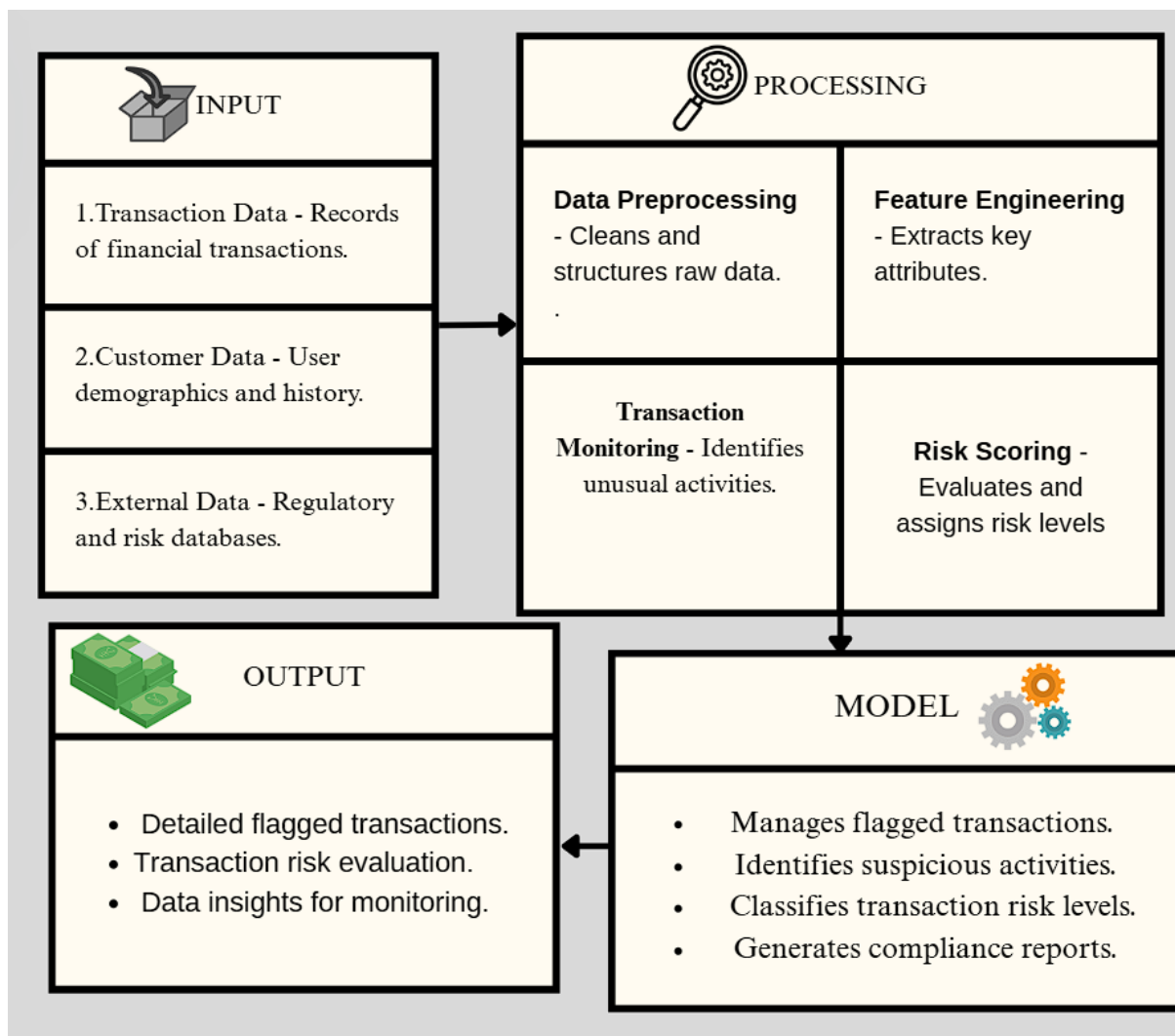


Fig 5.3.1 System Architecture.

The given AML system architecture in figure 5.3.1 consists of four main sections: Input, Processing, Model, and Output. Each of these sections plays a crucial role in identifying and flagging suspicious financial activities.

The Input section gathers data from multiple sources, which are essential for analyzing transactions and identifying potential money laundering activities. This includes transaction data, which contains records of all financial transactions, customer data, which holds user demographics, account details, and past transactions, and external data, which consists of regulatory and risk databases that help ensure compliance with legal frameworks.

The Processing stage prepares and analyzes the collected data to extract useful insights. First, data preprocessing is performed to clean and structure raw data, removing inconsistencies, missing values, and redundant entries. Then, feature engineering extracts key transaction attributes such as transaction amount, frequency, and recipient details. The system then uses transaction monitoring to identify unusual patterns, such as sudden large transfers or transactions involving high-risk countries. Simultaneously, risk scoring assigns a numerical risk level to each transaction based on machine learning models and past fraud trends.

Once the data is processed, the Model plays a central role in analyzing flagged transactions and generating insights. The model manages flagged transactions by identifying suspicious activities based on predefined patterns and AI-driven anomaly detection. It then classifies transaction risk levels, categorizing them into low, medium, or high-risk groups. Additionally, the model generates compliance reports to be shared with authorities.

5.4. Implementation:

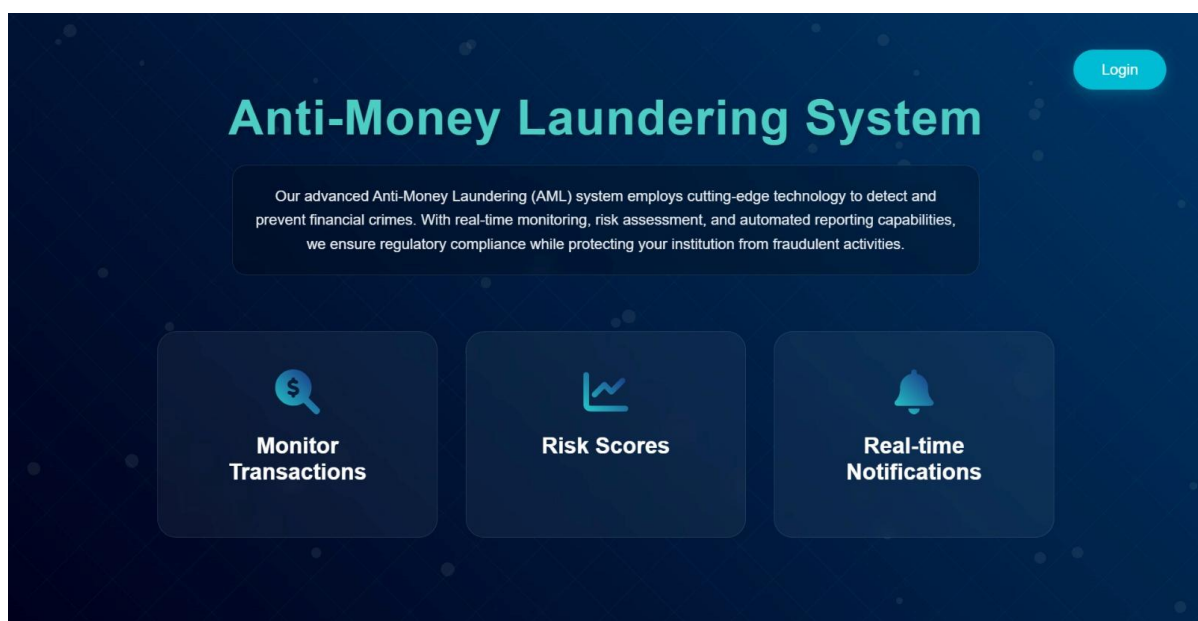
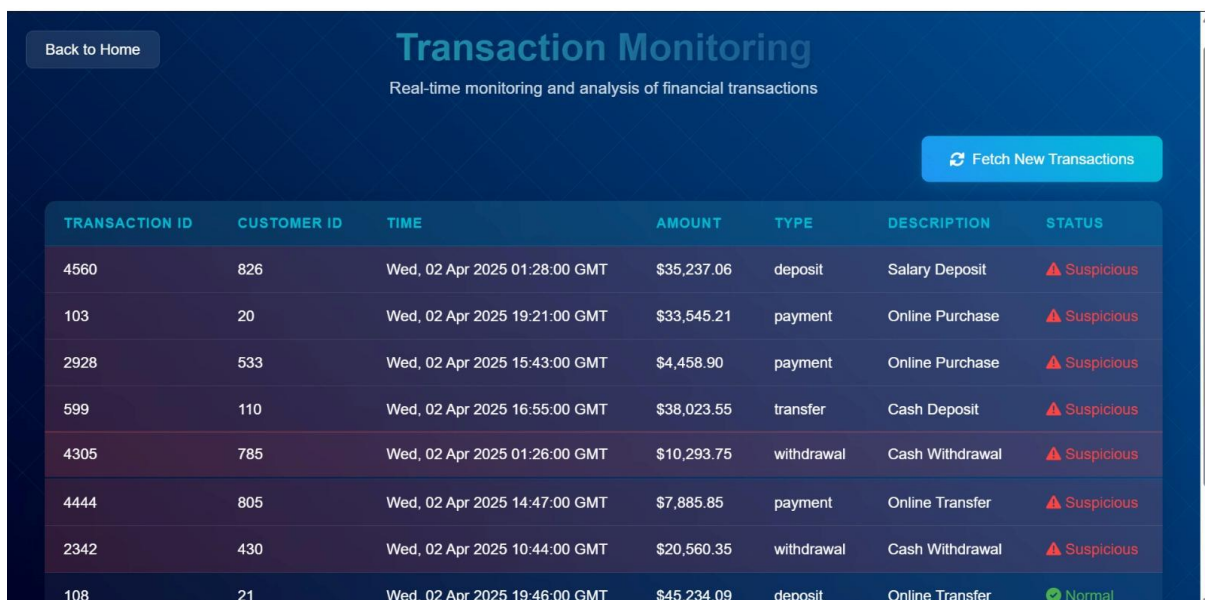


Fig 5.4.1 Dashboard

Above Figure 5.4.1 showcases the Anti-Money Laundering (AML) System Home Page, featuring a modern and professional dark-themed design. The system name, "Anti-Money Laundering System", is prominently displayed in a bold, blue-green gradient font, reinforcing its identity as a secure and technology-driven platform. A concise tagline highlights its advanced capabilities in real-time monitoring, risk assessment, and automated fraud detection, ensuring financial security and regulatory compliance. The homepage encourages users to engage with its key functionalities through three distinct feature cards—Monitor Transactions, Risk Scores, and Real-time Notifications—each visually represented with minimalist icons for clarity. A Login button is positioned at the top-right corner, providing secure access. The well-structured layout and intuitive navigation ensure a seamless user experience, making fraud detection more accessible and efficient.



TRANSACTION ID	CUSTOMER ID	TIME	AMOUNT	TYPE	DESCRIPTION	STATUS
4560	826	Wed, 02 Apr 2025 01:28:00 GMT	\$35,237.06	deposit	Salary Deposit	▲ Suspicious
103	20	Wed, 02 Apr 2025 19:21:00 GMT	\$33,545.21	payment	Online Purchase	▲ Suspicious
2928	533	Wed, 02 Apr 2025 15:43:00 GMT	\$4,458.90	payment	Online Purchase	▲ Suspicious
599	110	Wed, 02 Apr 2025 16:55:00 GMT	\$38,023.55	transfer	Cash Deposit	▲ Suspicious
4305	785	Wed, 02 Apr 2025 01:26:00 GMT	\$10,293.75	withdrawal	Cash Withdrawal	▲ Suspicious
4444	805	Wed, 02 Apr 2025 14:47:00 GMT	\$7,885.85	payment	Online Transfer	▲ Suspicious
2342	430	Wed, 02 Apr 2025 10:44:00 GMT	\$20,560.35	withdrawal	Cash Withdrawal	▲ Suspicious
108	21	Wed, 02 Apr 2025 19:46:00 GMT	\$45,234.09	deposit	Online Transfer	● Normal

Fig 5.4.2 Transaction Monitoring Page

Above Figure 5.4.2 showcases the Transaction Monitoring Page of the Anti-Money Laundering (AML) System, designed for real-time tracking and analysis of financial transactions. The page features a structured data table displaying key transaction details, including Transaction ID, Customer ID, Time, Amount, Type, Description, and Status. Suspicious transactions are marked with a red Suspicious label, while normal transactions are indicated with a green Normal tag, helping users quickly identify potentially fraudulent activities. A "Fetch New Transactions" button is prominently placed for refreshing the data in

real time, ensuring up-to-date monitoring. Additionally, a "Back to Home" button provides easy navigation.

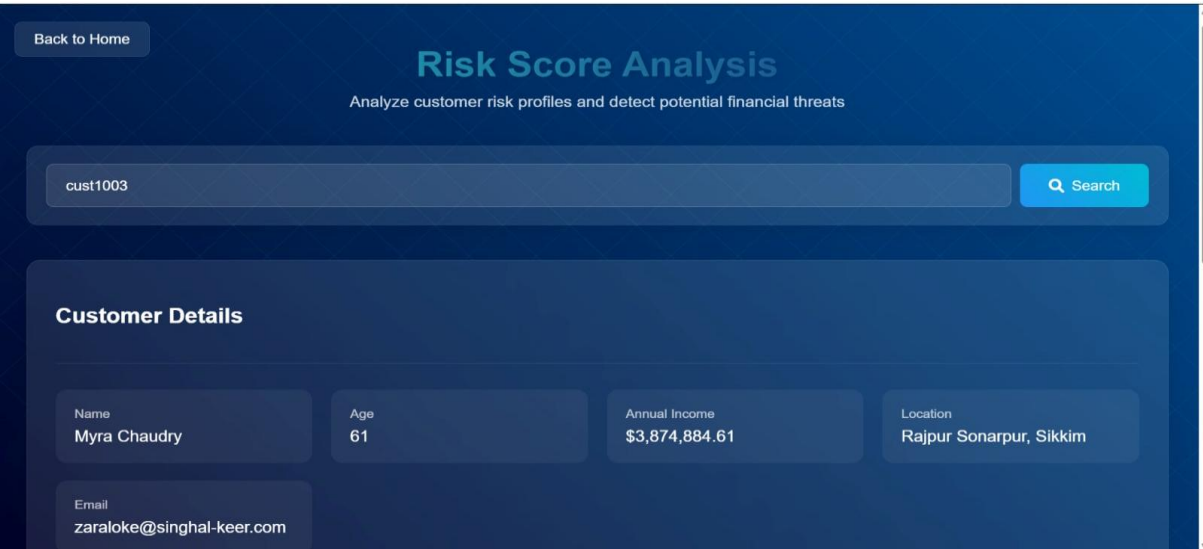


Fig 5.4.3 Risk Score Analysis Page

Figure 5.4.3 showcases the Risk Score Analysis Page of the Anti-Money Laundering (AML) System, designed to evaluate customer risk profiles and detect potential financial threats. The interface features a search bar, allowing users to input a customer ID and retrieve detailed risk-related information. Below, the Customer Details section displays essential data, including Name, Age, Annual Income, Email, and Location, providing a comprehensive overview of the user's financial background. The "Search" button enables quick retrieval of data, ensuring efficiency in risk assessment. The "Back to Home" button ensures seamless navigation.

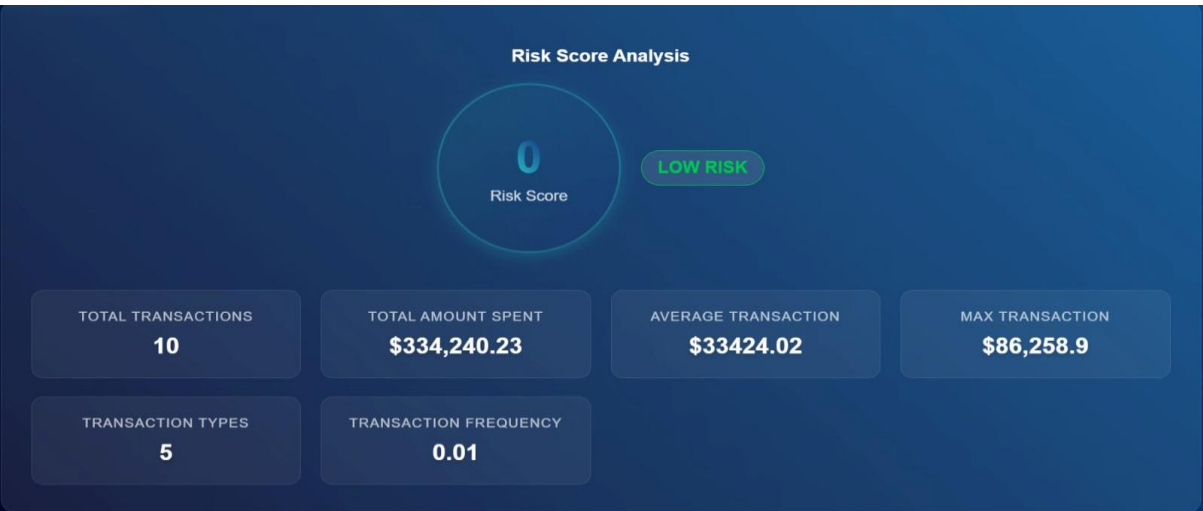


Fig 5.4.4 Risk Score Dashboard

Figure 5.4.4 showcases the Risk Score Analysis Dashboard of the Anti-Money Laundering (AML) System, designed to assess customer risk profiles and detect potential financial threats. The interface features a search bar for inputting a customer ID, retrieving detailed financial data efficiently. The Customer Details section displays key information such as Name, Age, Annual Income, Email, and Location, offering a complete overview of the customer's financial background. The Risk Score Analysis panel presents the Risk Score, categorizing the user as Low, Medium, or High Risk, along with key metrics like Total Transactions, Total Amount Spent, Average Transaction Value, Maximum Transaction, Transaction Types, and Transaction Frequency to provide deeper insights into transaction behavior. The structured dark-themed UI enhances readability and usability for financial analysts, ensuring seamless navigation with the Search and Back to Home buttons.

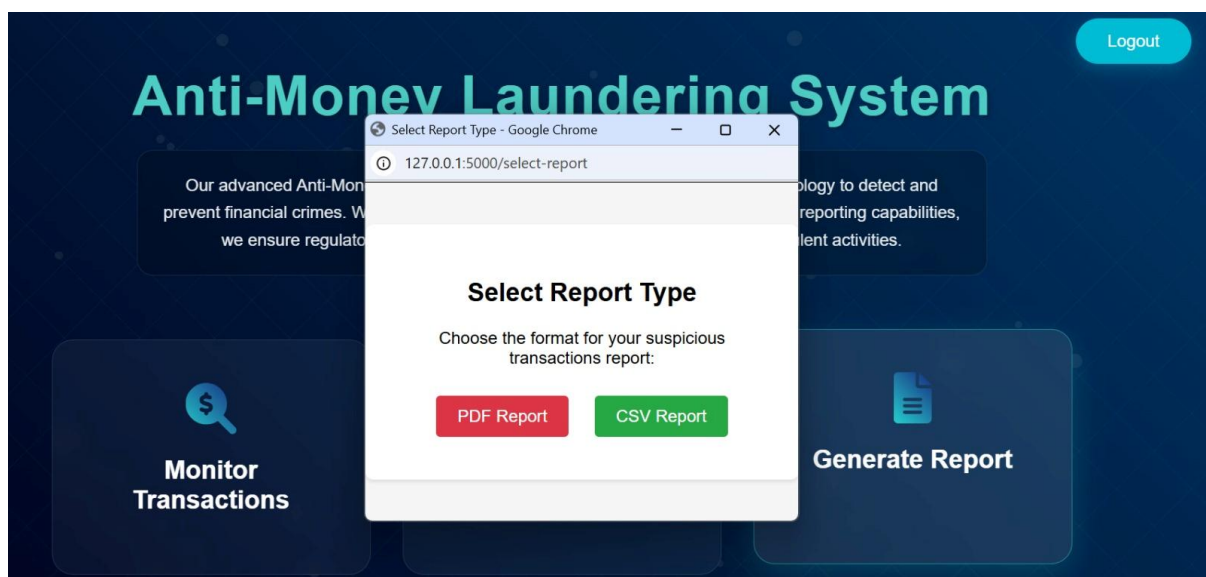


Fig 5.4.5 Report Generation

Figure 5.4.5 displays the Report Generation Interface of the Anti-Money Laundering (AML) System, enabling users to generate reports on suspicious transactions. The pop-up window titled "Select Report Type" prompts users to choose between two formats: PDF Report or CSV Report, ensuring flexibility in data export. The intuitive design enhances usability, with clear red and green buttons for selection. In the background, the dashboard highlights key AML system functionalities, such as Monitoring Transactions and Generating Reports, reinforcing the system's role in financial crime prevention.

Transaction ID	Customer ID	Time	Amount	Type	Description
4039	733	2025-04-02 23:53:00	\$4,085.88	transfer	Bank Transfer
5330	965	2025-04-02 22:43:00	\$41,957.90	transfer	Bank Transfer
4798	869	2025-04-02 22:29:00	\$4,086.74	payment	Restaurant Bill
1169	211	2025-04-02 22:08:00	\$2,360.47	transfer	Cash Deposit
382	71	2025-04-02 20:58:00	\$3,762.06	payment	Online Purchase
122	23	2025-04-02 20:49:00	\$19,969.14	withdrawal	ATM Withdrawal
5308	961	2025-04-02 20:36:00	\$8,797.91	payment	Cash Deposit
5038	908	2025-04-02 20:02:00	\$2,130.24	withdrawal	Cash Withdrawal
5299	960	2025-04-02 19:38:00	\$3,165.51	deposit	Online Transfer
5032	907	2025-04-02 19:37:00	\$4,510.99	withdrawal	Cash Withdrawal
627	114	2025-04-02 19:26:00	\$1,124.44	payment	Online Transfer
3728	678	2025-04-02 18:59:00	\$1,800.39	withdrawal	ATM Withdrawal
3979	721	2025-04-02 18:34:00	\$38,029.42	payment	Online Purchase
3978	721	2025-04-02 18:28:00	\$23,880.98	deposit	Check Deposit
2705	494	2025-04-02 18:11:00	\$2,449.62	payment	Online Purchase
347	64	2025-04-02 18:02:00	\$1,267.34	transfer	Online Transfer
3457	624	2025-04-02 17:07:00	\$12,694.22	transfer	Online Transfer
334	61	2025-04-02 16:39:00	\$1,388.13	withdrawal	ATM Withdrawal
2686	491	2025-04-02 16:30:00	\$8,057.04	deposit	Cash Deposit
4210	765	2025-04-02 16:15:00	\$31,195.39	withdrawal	ATM Withdrawal
1633	296	2025-04-02 16:01:00	\$2,076.48	transfer	Online Transfer
4984	900	2025-04-02 15:24:00	\$49,587.02	transfer	Online Transfer

Fig 5.4.6 Suspicious transactions report

Figure 5.4.6 presents the Suspicious Transactions Report from the Anti-Money Laundering (AML) System, displaying a structured table of flagged financial transactions. The table includes key details such as Transaction ID, Customer ID, Timestamp, Amount, Transaction Type, and Description, enabling analysts to assess potentially fraudulent activities. Transactions cover various categories like Bank Transfers, ATM Withdrawals, Online Purchases, and Cash Deposits, with high-value transactions highlighted for further investigation. The clean, grid-based format enhances readability, supporting financial analysts in identifying unusual patterns and mitigating risks effectively.

Chapter 6

Technical Specification

Technology specification refers to a detailed outline of the technologies, tools, and methodologies that will be used in a project. It encompasses the selection of programming languages, frameworks, libraries, and platforms that will shape the development process and define how various components of the system interact. The specification serves as a guideline for the development team, ensuring consistency and coherence throughout the project lifecycle.

Frontend:

The frontend of this project is developed using HTML5, CSS, and JavaScript, ensuring an interactive and user-friendly interface.

- HTML5 is used for structuring the web pages, defining the layout of different sections such as user dashboards, transaction records, risk score visualizations, and reports. It also supports embedding interactive charts and forms for user input.
- CSS is utilized for styling and designing the interface, ensuring a visually appealing and responsive layout. The use of CSS Flexbox and Grid allows for proper alignment of elements, while media queries ensure adaptability across different screen sizes.
- JavaScript enhances the interactivity of the application, enabling dynamic updates without reloading the page. It handles features such as real-time transaction data visualization, filtering of risk scores, and event-based updates.

Backend:

The backend is built using Flask, a lightweight Python web framework that handles the logic for processing user requests, managing transactions, and risk score computation. Flask enables seamless integration with machine learning models, database operations, and API handling.

- The Flask application serves RESTful APIs for fetching and processing transaction data dynamically.
- It manages user authentication, ensuring secure access to risk assessments and reports.

- Flask also facilitates interaction with the database, executing queries efficiently and providing real-time data updates.

Algorithms:

1. Isolation Forest:-

Isolation Forest is an unsupervised learning algorithm designed for anomaly detection. It works by randomly selecting features and splitting data points recursively, isolating anomalies in fewer steps since they differ significantly from normal data. This makes it highly efficient for large datasets and useful in fraud detection, cybersecurity, and medical diagnostics. Unlike traditional clustering-based methods, Isolation Forest does not rely on density estimation, making it faster and scalable.

2. Random Forest:-

Random Forest, on the other hand, is a supervised learning algorithm used for classification and regression. It constructs multiple decision trees using random subsets of data and features, then aggregates their outputs through majority voting (classification) or averaging (regression). This ensemble method reduces overfitting and improves accuracy, making it widely used in finance, healthcare, stock market predictions, and recommendation systems. While Isolation Forest focuses on detecting outliers, Random Forest is a general-purpose algorithm for making predictions based on structured data.

Several Python libraries are integrated into the project for efficient data processing, machine learning, and report generation:

- **scikit-learn:** Used for implementing machine learning models, including Isolation Forest and Random Forest.
- **pandas:** Handles data manipulation and transaction record analysis.
- **numpy:** Supports numerical computations required for processing large datasets.
- **matplotlib:** Used for visualizing risk scores, transaction patterns, and anomalies.
- **reportlab:** Generates PDF reports summarizing transaction risks and user assessments.

Chapter 7

Project Schedule

A Gantt chart is a visual project management tool that provides a clear timeline of tasks, their duration, and dependencies. It helps in tracking project progress, ensuring efficient task allocation, and meeting deadlines. In this project, the Gantt chart was used to outline key milestones, responsibilities, and deadlines across two major phases: Project Conception and Initiation and Project Design and Implementation. This structured approach ensured smooth execution, timely completion, and effective collaboration among team members.

It maps out each task with its corresponding start and end dates, duration, and dependencies, giving a clear overview of how different components of the project align. In this project, the Gantt chart was used to plan and organize all major and minor tasks from the initial stages to final completion. It helped in identifying task sequences, prioritizing actions, and allocating responsibilities to individual team members. The Gantt chart thus acted as a living document that guided the team through the project lifecycle, fostering collaboration and ensuring that deliverables were met within the stipulated timeframe. The detailed breakdown of tasks and their completion status is presented in the following Gantt chart.

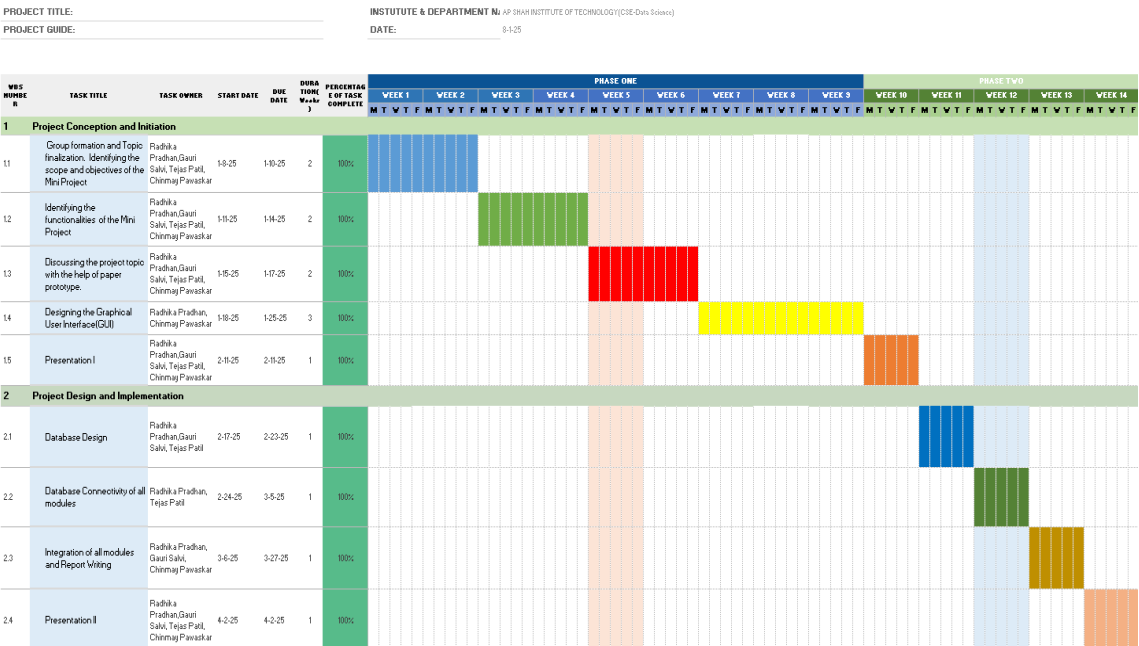


Fig 7.1 Gantt Chart

Following is the detail of the Gantt chart – The project was structured into two main phases: Project Conception and Initiation and Project Design and Implementation, spanning a total of 14 weeks. The first phase began with group formation and topic finalization in Week 1, where the team, consisting of Radhika, Pradhan, Gauri Salvi, Tejas Patil, and Chinmay Pawaskar, identified the project's scope and objectives. In Week 2, the team outlined the functionalities of the mini-project, ensuring feasibility and technical requirements were met. This was followed by paper prototype discussions in Week 3 to visualize the project's workflow. The Graphical User Interface (GUI) design was initiated in Week 4 and finalized in Week 5 after incorporating necessary improvements. The first phase concluded with Presentation I in Week 6, where the team showcased their progress.

The second phase, Project Design and Implementation, commenced in Week 7 with database design, where the team planned and structured the database schema. By Week 8, the database structure was implemented, and in Week 9, database connectivity with all modules was successfully established. Weeks 10 and 11 were dedicated to module integration and report writing, ensuring seamless interaction between different components of the system. In Weeks 12 and 13, extensive testing and debugging were performed, followed by the preparation of the final project report and presentation. The project concluded in Week 14 with Presentation II, where the team demonstrated the fully functional system.

Chapter 8

Results

The implementation of the AI-driven Anti-Money Laundering (AML) System is set to revolutionize financial security by utilizing advanced machine learning algorithms and real-time transaction monitoring. The anticipated results include enhanced fraud detection accuracy, enabling financial institutions to identify and flag suspicious activities with greater precision. Additionally, the system ensures efficient risk assessment, providing automated risk scores for customers based on transaction patterns. With a user-friendly dashboard and real-time analytics, analysts can quickly detect anomalies, track suspicious transactions, and make data-driven decisions, significantly improving financial security and operational efficiency.

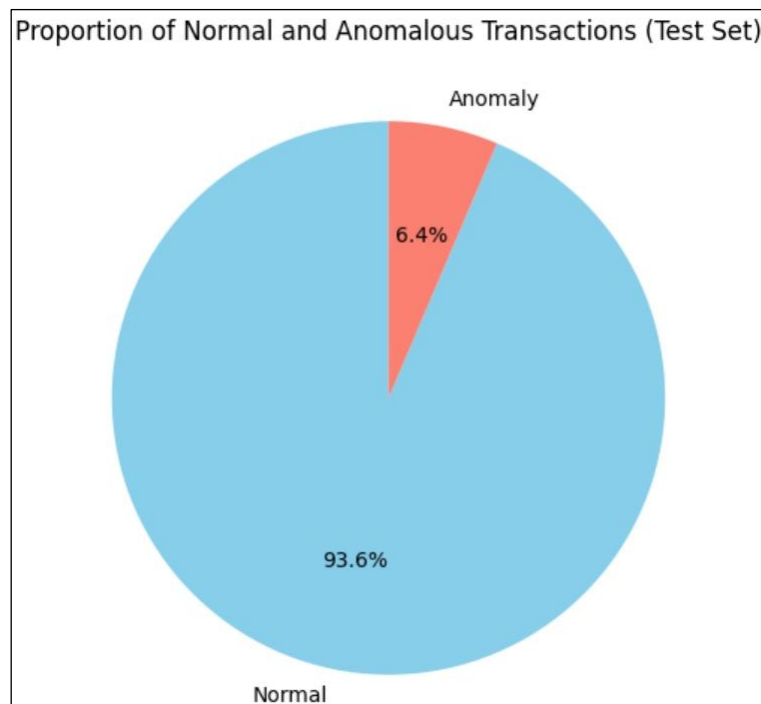


Fig 8.1 Isolation Forest Anomaly Detection Results

The pie chart in the figure 8.1, illustrates the results of an Isolation Forest model applied to a test dataset for fraud detection, categorizing transactions as either normal or anomalous. The analysis reveals that 93.6% of transactions are classified as normal, indicating typical financial activities, while 6.4% are flagged as anomalies, suggesting potential fraudulent or suspicious transactions. This result demonstrates the model's ability to isolate rare, irregular patterns in

financial data, which could help financial institutions in identifying high-risk transactions for further investigation.

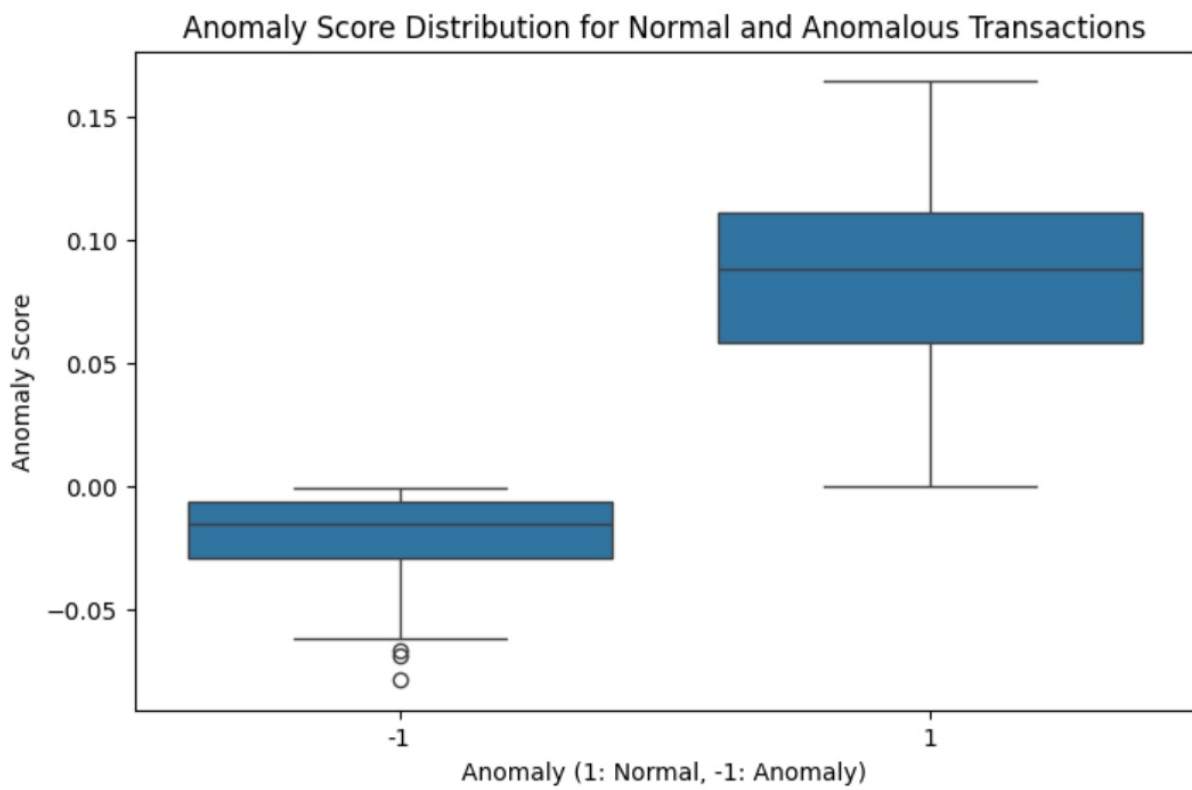


Fig 8.2 Anomaly Score Distribution for Normal and Anomalous Transactions

The box plot in the figure 8.2, illustrates the results of an Isolation Forest model applied to a test dataset for fraud detection, showing the distribution of anomaly scores for normal and anomalous transactions. The analysis reveals a clear distinction between the two categories: normal transactions (labeled as 1) have lower anomaly scores, indicating consistent financial behavior, while anomalous transactions (labeled as -1) exhibit significantly higher anomaly scores, suggesting potential fraudulent activity. This result demonstrates the model's effectiveness in identifying unusual patterns in financial transactions, helping financial institutions detect high-risk activities for further investigation.

Accuracy: 0.979381443298969					
Classification Report:					
		precision	recall	f1-score	support
	0	0.99	0.99	0.99	102
	1	0.96	0.98	0.97	87
	2	0.99	0.97	0.98	102
	accuracy			0.98	291
	macro avg	0.98	0.98	0.98	291
	weighted avg	0.98	0.98	0.98	291

Fig 8.3 Classification report of random forest

The classification report shown in figure 8.3, indicates a highly accurate machine learning model with an overall accuracy of 97.94%. The model demonstrates strong precision, recall, and F1-scores across all three classes (0, 1, and 2), suggesting minimal false positives and effective detection of actual instances. The recall values highlight the model's ability to correctly identify each class, while the F1-scores, nearing 1, confirm a balanced performance. The macro and weighted averages remain at 0.98, further reinforcing the model's reliability in distinguishing between different transaction categories. With a strong performance across all metrics, this model is well-suited for tasks such as fraud detection or anomaly classification in financial transactions.

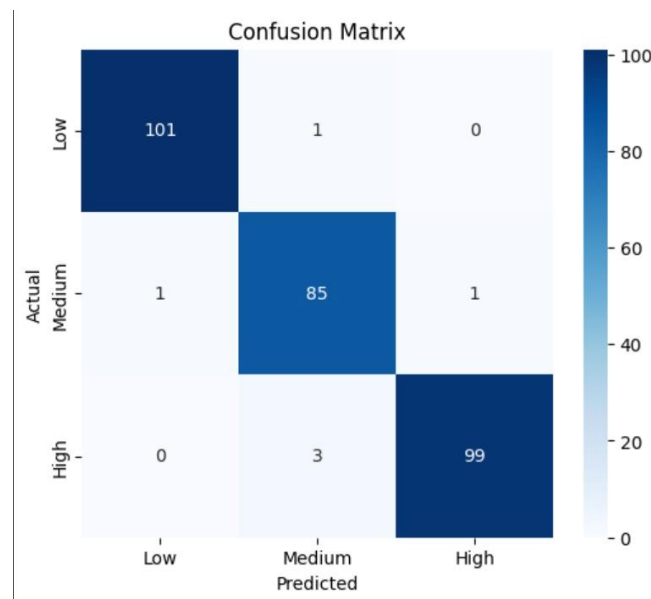


Fig 8.4 Confusion Matrix

The figure 8.4 shows a confusion matrix which illustrates the performance of a multi-class classification model in predicting three categories: Low, Medium, and High. The model demonstrates strong accuracy, correctly classifying most instances within their respective categories. For the "Low" class, 101 transactions were correctly classified, with only 1 misclassified as "Medium" and none as "High." Similarly, for the "Medium" class, 85 were correctly identified, but 1 was misclassified as "Low," and another as "High." The "High" category shows 99 correct predictions, with 3 misclassified as "Medium" but none as "Low." These results suggest that the model performs well, with minimal misclassifications, particularly between Medium and High, where slight overlaps occur.

Chapter 9

Conclusion

In conclusion, the AI-driven Anti-Money Laundering (AML) system enhances financial security by leveraging advanced machine learning techniques such as Isolation Forest for transaction monitoring and Random Forest for risk score generation. By detecting anomalies in financial transactions through Isolation Forest, the system efficiently identifies suspicious activities with minimal false positives. Simultaneously, the use of Random Forest ensures accurate and dynamic risk scoring, enabling financial institutions to assess and prioritize potential threats effectively. This combination not only improves fraud detection accuracy but also enhances regulatory compliance by providing explainable and data-driven insights. As the system continues to evolve, integrating real-time monitoring, adaptive learning models, and secure reporting mechanisms, it will further strengthen financial crime prevention and ensure a more resilient AML framework.

Chapter 10

Future Scope

The future scope of AI-driven Anti-Money Laundering (AML) systems is vast, offering numerous possibilities for enhancement and innovation. As financial institutions accumulate transaction data, machine learning models can be continuously improved to increase the accuracy and efficiency of fraud detection. Potential advancements include the integration of advanced deep learning techniques and explainable AI (XAI) to enhance transparency in risk assessments, enabling regulators and financial analysts to better understand AI-driven decisions. Additionally, real-time monitoring and adaptive risk scoring mechanisms will help detect suspicious activities more proactively, minimizing financial crimes and regulatory violations.

Furthermore, AML systems can evolve by incorporating blockchain technology to enhance transaction traceability and reduce fraudulent activities across global financial networks. Enhanced collaboration between banks, regulatory bodies, and law enforcement agencies through secure AI-driven information-sharing platforms will strengthen fraud detection capabilities. The expansion of AI-driven AML solutions to decentralized finance (DeFi) and cryptocurrency transactions will further ensure financial security in the evolving digital economy. Additionally, integrating natural language processing (NLP) for automated report generation and compliance checks can streamline regulatory processes, reducing manual workload. Ultimately, by embracing these advancements, AI-powered AML systems can significantly strengthen financial security, mitigate risks, and adapt to the ever-changing landscape of financial crime.

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